

The Grid-Minor Theorem Revisited*

Vida Dujmović[†] Robert Hickingbotham[‡] Jędrzej Hodor[§] Gwenaël Joret[¶]
 Hoang La^{||} Piotr Micek^{**} Pat Morin^{††} Clément Rambaud^{‡‡} David R. Wood^{§§}

Abstract

We prove that for every planar graph X of treedepth h , there exists a positive integer c such that for every X -minor-free graph G , there exists a graph H of treewidth at most $f(h)$ such that G is isomorphic to a subgraph of $H \boxtimes K_c$. This is a qualitative strengthening of the Grid-Minor Theorem of Robertson and Seymour (JCTB, 1986), and treedepth is the optimal parameter in such a result. As an example application, we use this result to improve the upper bound for weak coloring numbers of graphs excluding a given graph as a minor.

The seminal *Graph Minors* series of Robertson and Seymour is the foundation of modern structural graph theory. In this work, treewidth is a central concept that measures how similar a given graph is to a tree. A key theorem of Robertson and Seymour [23] states that a minor-closed graph class $\mathcal{G}$ has bounded treewidth if and only if some planar graph is not in $\mathcal{G}$. In particular, for every planar graph X , every X -minor-free graph has treewidth at most some function $g(X)$. This result is often called the Grid-Minor Theorem since it suffices to prove it when X is a planar grid. The asymptotics of g have been substantially improved since the original work. Most significantly, Chekuri and Chuzhoy [4] showed that g can be chosen to be polynomial in $|V(X)|$. The current best bound is $g(X) \in \tilde{O}(|V(X)|^9)$, which follows from a result of Chuzhoy and Tan [5]. Dependence on $|V(X)|$ is unavoidable, since the complete graph on $|V(X)| - 1$ vertices is X -minor-free, but has treewidth $|V(X)| - 2$. Our goal is to prove a qualitative strengthening of the Grid-Minor Theorem via graph product structure theory.

Graph product structure theory describes graphs in complicated classes as subgraphs of strong products¹ of simpler graphs. For example, the Planar Graph Product Structure Theorem by Dujmović, Joret, Micek, Morin, Ueckerdt, and Wood [8] says that for every planar graph G there is a graph H of treewidth at most 3 and a path P such that $G \subseteq H \boxtimes P \boxtimes K_3$. Here, $G_1 \subseteq G_2$ means that G_1 is isomorphic to a subgraph of G_2 .

*The full version of the paper is available at <https://arxiv.org/abs/2307.02816>.

[†]School of Computer Science and Electrical Engineering, University of Ottawa, Ottawa, Canada (vida.dujmovic@uottawa.ca). Research supported by NSERC, and a Gordon Preston Fellowship from the School of Mathematics at Monash University.

[‡]School of Mathematics, Monash University, Melbourne, Australia (robert.hickingbotham@monash.edu). Research supported by an Australian Government Research Training Program Scholarship.

[§]Theoretical Computer Science Department, Faculty of Mathematics and Computer Science, Jagiellonian University, Kraków, Poland (jedrzej.hodor@gmail.com).

[¶]Département d'Informatique, Université libre de Bruxelles, Belgium (gwenael.joret@ulb.be). G. Joret is supported by a PDR grant from the Belgian National Fund for Scientific Research (FNRS), and by the Australian Research Council.

^{||}Theoretical Computer Science Department, Faculty of Mathematics and Computer Science, Jagiellonian University, Kraków, Poland (hoang.la.research@gmail.com).

^{**}Theoretical Computer Science Department, Faculty of Mathematics and Computer Science, Jagiellonian University, Kraków, Poland (piotr.micek@uj.edu.pl). Hodor, La and Micek are supported by the National Science Centre of Poland under grant 2018/31/G/ST1/03718 within the BEETHOVEN program.

^{††}School of Computer Science, Carleton University, Ottawa, Canada (morin@scs.carleton.ca). Research supported by NSERC.

^{‡‡}DIENS, École Normale Supérieure, CNRS, PSL University, Paris, France (clement.rambaud@inria.fr).

^{§§}School of Mathematics, Monash University, Melbourne, Australia (david.wood@monash.edu). Research supported by the Australian Research Council.

¹The *strong product* $G_1 \boxtimes G_2$ of two graphs G_1 and G_2 is the graph with vertex set $V(G_1 \boxtimes G_2) := V(G_1) \times V(G_2)$ and that includes the edge with endpoints (v, x) and (w, y) if and only if $vw \in E(G_1)$ and $x = y$; $v = w$ and $xy \in E(G_2)$; or $vw \in E(G_1)$ and $xy \in E(G_2)$.

Inspired by this viewpoint, we prove the following product structure extension of the Grid-Minor Theorem. Note that $H \boxtimes K_c$ is the graph obtained from H by ‘blowing-up’ each vertex of H by a complete graph K_c . Let $\text{tw}(G)$ denote the treewidth¹ of a graph G , and let $\text{td}(G)$ denote the treedepth² of G .

THEOREM 1. *For every planar graph X , there exists a positive integer c such that for every X -minor-free graph G , there exists a graph H of treewidth at most $2^{\text{td}(X)+1} - 4$ such that $G \subseteq H \boxtimes K_c$.*

The point of Theorem 1 is that the treewidth of H only depends on the treedepth of X , not on $|V(X)|$ ³. The described product structure of G is a more refined description of G compared to the output of the Grid-Minor Theorem since $\text{tw}(H \boxtimes K_c) \leq (\text{tw}(H) + 1)c - 1$. This refinement is useful because various graph parameters can be bounded on $H \boxtimes K_c$ by a fast-growing function of $\text{tw}(H)$ times a slow-growing (usually linear) function of c ; this includes queue-number [8], nonrepetitive chromatic number [9], and others [2, 12]. As concrete applications of Theorem 1, we use it to improve bounds for weak coloring numbers and p -centered colorings of X -minor-free graphs.

The Grid-Minor Theorem relates to treewidth in the same way as the Excluded-Tree-Minor Theorem by Robertson and Seymour [22] relates to pathwidth. The latter says that a minor-closed class $\mathcal{G}$ has bounded pathwidth if and only if some tree is not in $\mathcal{G}$. In particular, there is a function g such that for every tree X , every X -minor-free graph has pathwidth at most $g(|V(X)|)$. The following product structure version of this result was proved by Dujmović, Hickingbotham, Joret, Micek, Morin, and Wood [10]: there exists a function f such that for every tree X , there exists a positive integer c such that for every X -minor-free graph G , there exists a graph H of pathwidth at most $f(\text{td}(X))$ such that $G \subseteq H \boxtimes K_c$. (In fact, they prove a stronger statement in which the pathwidth of H is bounded by $2h - 1$, where h is the radius of X .)

We actually prove the following result for an arbitrary excluded minor, which combined with the Grid-Minor Theorem immediately implies Theorem 1.

THEOREM 2. *For every graph X , there exists a positive integer c such that for every positive integer t and for every X -minor-free graph G with $\text{tw}(G) < t$, there exists a graph H of treewidth at most $2^{\text{td}(X)+1} - 4$ such that $G \subseteq H \boxtimes K_{ct}$.*

Theorem 2 was inspired by and can be restated in terms of the parameter ‘underlying treewidth’ introduced by Campbell, Clinch, Distel, Gollin, Hendrey, Hickingbotham, Huynh, Illingworth, Tamitegama, Tan, and Wood [3]. They defined the *underlying treewidth* of a graph class $\mathcal{G}$, denoted by $\text{utw}(\mathcal{G})$, to be the smallest integer such that, for some function $f : \mathbb{N} \rightarrow \mathbb{N}$, for every graph $G \in \mathcal{G}$ there is a graph H of treewidth at most $\text{utw}(\mathcal{G})$ such that $G \subseteq H \boxtimes K_{f(\text{tw}(G))}$. Here, f is called the *treewidth binding function*. Campbell et al. [3] showed that the underlying treewidth of the class of planar graphs equals 3, and the same holds for any fixed surface. More generally, let $\mathcal{G}_X$ be the class of graphs excluding a given graph X as a minor. Campbell et al. [3] showed that $\text{utw}(\mathcal{G}_{K_t}) = t - 2$ and $\text{utw}(\mathcal{G}_{K_{s,t}}) = s$ (for $t \geq \max\{s, 3\}$). In these results, the treewidth binding function is quadratic. Illingworth, Scott and Wood [16] reproved these results with a linear treewidth binding function.

Determining the underlying treewidth of the class of X -minor-free graphs, for an arbitrary graph X , was one of the main problems left unsolved by Campbell et al. [3] and Illingworth, Scott and Wood [16]. Theorem 2 together with a well-known lower bound construction shows that $\text{utw}(\mathcal{G}_X)$ and $\text{td}(X)$ are tied:

$$\text{td}(X) - 2 \leq \text{utw}(\mathcal{G}_X) \leq 2^{\text{td}(X)+1} - 4.$$

¹A *tree decomposition* $\mathcal{W}$ of a graph G is a pair $(T, (W_x \mid x \in V(T)))$, where T is a tree and for each $x \in V(T)$ the set W_x is a subset of $V(G)$ called a *bag*, such that (a) for each edge $uv \in E(G)$ there is a bag containing both u and v , and (b) for each vertex $v \in V(G)$ the set of vertices $x \in V(T)$ with $v \in W_x$ induces a non-empty subtree of T . The *width* of $\mathcal{W}$ is $\max\{|W_x| - 1 \mid x \in V(T)\}$, and its *adhesion* is $\max\{|W_x \cap W_y| \mid xy \in E(T)\}$. The *treewidth* of a graph G , denoted $\text{tw}(G)$, is the minimum width of a tree decomposition of G .

²A *rooted forest* is a disjoint union of rooted trees. The *vertex-height* of a rooted forest F is the maximum number of vertices on a path from a root to a leaf in F . For two vertices u, v in a rooted forest F , we say that u is a *descendant* of v in F if v lies on the path from a root to u in F . Note that v is a descendant of itself; descendants of v that are distinct from v are called *strict descendants* of v . The *closure* of F is the graph with vertex set $V(F)$ and edge set $\{vw \mid v \text{ is a strict descendant of } w \text{ in } F\}$. The *treedepth* of a graph G , denoted by $\text{td}(G)$, is 0 if G is empty, and otherwise is the minimum vertex-height of a rooted forest F with $V(F) = V(G)$ such that G is a subgraph of the closure of F .

³Arbitrarily large graphs can have bounded treedepth, such as edgeless graphs (treedepth 1) or stars (treedepth 2).

This shows that treedepth is the right parameter to consider in Theorems 1 and 2. Moreover, in the upper bound the treewidth binding function is linear.

Application #1. Weak Coloring Numbers. Weak coloring numbers are a family of graph parameters studied extensively in structural and algorithmic graph theory. See the book by Nešetřil and Ossona de Mendez [18], or the recent lecture notes by Pilipczuk, Pilipczuk, and Siebertz [19] for more information on this topic. For the algorithmic side, see Dvořák [11] and also Theorem 5.2 in [19], which contains a polynomial-time approximation algorithm for r -dominating set, with approximation ratio bounded by a function of a weak coloring number of the input graph. We now introduce the definition. The *length* of a path is the number of its edges. For two vertices u and v in a graph G , a u - v *path* is a path in G with endpoints u and v . Let G be a graph and let σ be an ordering of the vertices of G . For an integer $r \geq 1$ and two vertices u and v of G , we say that u is *weakly r -reachable* from v in σ , if there exists a u - v path of length at most r such that for every vertex w on the path, $u \leq_\sigma w$. The set of vertices that are weakly r -reachable from a vertex v in σ is denoted by $\text{WReach}_r[G, \sigma, v]$. The r -th *weak coloring number* of G , denoted by $\text{wcol}_r(G)$, is defined as

$$\text{wcol}_r(G) = \min_{\sigma} \max_{v \in V(G)} |\text{WReach}_r[G, \sigma, v]|.$$

where σ ranges over the set of all vertex orderings of G . Several papers give bounds for weak coloring numbers of graphs in a given class of sparse graphs. For example, if G is planar, then $\text{wcol}_r(G) = \mathcal{O}(r^3)$; if G has no K_t -minor, then $\text{wcol}_r(G) = \mathcal{O}(r^{t-1})$ as proved by van den Heuvel, Ossona de Mendez, Quiroz, Rabinovich, and Siebertz [15]; and if $\text{tw}(G) \leq t$, then $\text{wcol}_r(G) \leq \binom{r+t}{t}$ and this bound is tight as proved by Grohe, Kreutzer, Rabinovich, Siebertz, and Stavropoulos [14].

Fix a planar graph X . What is known about $\text{wcol}_r(G)$ when X is not a minor of G ? Since G is $K_{|V(X)|}$ -minor-free, $\text{wcol}_r(G) = \mathcal{O}(r^{|V(X)|-1})$. However, thanks to Theorem 1, there exists a graph H with $\text{tw}(H) \leq f(\text{td}(X)) = \mathcal{O}(2^{\text{td}(X)})$ and c depending only on X such that $G \subseteq H \boxtimes K_c$. Therefore,

$$\text{wcol}_r(G) \leq \text{wcol}_r(H \boxtimes K_c) \leq c \cdot \text{wcol}_r(H) \leq c \cdot r^{f(\text{td}(X))}.$$

Indeed, the first inequality follows from the monotonicity of wcol_r and the second inequality is an easy property¹ of wcol_r . The obtained upper bound on $\text{wcol}_r(G)$ is polynomial in r , where the exponent depends only on $\text{td}(X)$ and not on $|V(X)|$.

As mentioned above, the Grid-Minor Theorem and also Theorem 1 only hold when the excluded minor X is planar. However, Theorem 2 does not have this restriction, hence there is no obvious obstacle for the above bound on $\text{wcol}_r(G)$ to hold for all graphs X . We prove that this is indeed the case, which is the second main contribution of this paper.

THEOREM 3. *There exists a function g such that for every graph X , there exists an integer c such that for every X -minor-free graph G and every positive integer r ,*

$$\text{wcol}_r(G) \leq c \cdot r^{g(\text{td}(X))}.$$

Again, the point of Theorem 3 is that the degree of the polynomial in r bounding $\text{wcol}_r(G)$ depends only on $\text{td}(X)$ and not on $|V(X)|$. In the previous best bound for weak colouring number of X -minor free graphs, the degree of the polynomial in r depended on the vertex cover number² $\tau(X)$. In particular, it follows from a result by van den Heuvel and Wood [26, Proposition 28] regarding the weak colouring number of $K_{s,t}^*$ -minor-free graphs that $\text{wcol}_r(G) \leq c \cdot r^{\tau(X)+1}$ for every X -minor-free graph G and integer $r \geq 1$. Theorem 3 is qualitatively stronger since $\text{td}(X) \leq \tau(X) + 1$ and there are graphs X with $\text{td}(X) = 3$ and arbitrarily large $\tau(X)$ (such as 1-subdivisions of stars).

Application #2. Product Structure for Apex-Minor-Free Graphs. As already mentioned, Dujmović et al. [8] proved that every planar graph is isomorphic to a subgraph of $H \boxtimes P \boxtimes K_3$ for some graph H

¹Let σ_H be an ordering of $V(H)$ such that $|\text{WReach}_r[H, \sigma_H, x]| \leq \text{wcol}_r(H)$ for every $x \in V(H)$. Consider σ an ordering of $V(H \boxtimes K_c)$ such that for every $(x, u), (x', u') \in V(H \boxtimes K_c)$ if $x <_{\sigma_H} x'$, then $(x, u) <_{\sigma} (x', u')$. It is easy to see that $\text{WReach}_r[H \boxtimes K_c, \sigma, (x, u)] \subseteq \{(y, v) \in V(H \boxtimes K_c) \mid y \in \text{WReach}_r[H, \sigma_H, x]\}$, and so $|\text{WReach}_r[H \boxtimes K_c, \sigma, u]| \leq c \cdot \text{wcol}_r(H)$.

²The *vertex cover number* $\tau(G)$ of a graph G is the size of a smallest set $S \subseteq V(G)$ such that every edge of G has at least one endpoint in S .

with treewidth at most 3 and for some path P . This result has been the key ingredient in the solution of several open problems on planar graphs [7–9, 13]. Building on this work, Distel, Hickingbotham, Huynh, and Wood [6] proved that every graph of Euler genus g is isomorphic to a subgraph of $H \boxtimes P \boxtimes K_{\max\{2g,3\}}$ for some graph H with treewidth at most 3 and for some path P . More generally, Dujmović et al. [8] characterized the graphs X for which there exist integers t and c such that every X -minor-free graph is isomorphic to a subgraph of $H \boxtimes P \boxtimes K_c$ where $\text{tw}(H) \leq t$ and P is a path. The answer is precisely the apex graphs. Here a graph X is *apex* if $V(X) = \emptyset$ or $X - u$ is planar for some vertex u of X . The following natural problem arises: for a given apex graph X , what is the minimum integer $t(X)$ such that, for some integer c , every X -minor-free graph is isomorphic to a subgraph of $H \boxtimes P \boxtimes K_c$ where $\text{tw}(H) \leq t(X)$ and P is a path? Illingworth, Scott and Wood [16] showed that $t(X) \leq \tau(X)$. We show, via an application of Theorem 2, that t is tied to treedepth. In particular,

$$(0.1) \quad \text{td}(X) - 2 \leq t(X) \leq 2^{\text{td}(X)+1} - 1.$$

Application #3. p -Centered Colorings. Theorem 1 can also be used to improve bounds for p -centered chromatic numbers of graphs excluding a fixed minor. For an integer $p \geq 1$, a vertex coloring ϕ of a graph G is *p -centered* if for every connected subgraph H of G either ϕ uses more than p colors in H or there is a color that appears exactly once in H . The *p -centered chromatic number* of G , denoted by $\chi_p(G)$, is the least number of colors in a p -centered coloring of G . p -Centered colourings are important since they characterize graph classes of bounded expansion [18], and are a central tool for designing parameterized algorithms in classes of bounded expansion [20, 21]; see [12] for an overview.

If $\text{tw}(G) \leq t$ then $\chi_p(G) \leq \binom{p+t}{t}$, and this bound is again tight [12, 20]. If X is a planar graph and G is X -minor-free, then by Theorem 1, there exists a graph H with $\text{tw}(H) \leq f(\text{td}(X))$ and c depending only on X such that $G \subseteq H \boxtimes K_c$. Therefore,

$$\chi_p(G) \leq \chi_p(H \boxtimes K_c) \leq c \cdot \chi_p(H) \leq c \cdot p^{f(\text{td}(X))},$$

where the first inequality follows from the monotonicity of χ_p and the second inequality is an easy property of p -centered colorings (Lemma 8 in [12]). The obtained upper bound on $\chi_p(G)$ is polynomial in p , where the exponent depends only on $\text{td}(X)$. Similarly, for an apex graph X , we use (0.1) to show that for some $c = c(X)$, every X -minor-free graph G satisfies $\chi_p(G) \leq c \cdot p^{f(\text{td}(X))}$.

Proof Ideas. This extended abstract omits all proofs. We now briefly mention some connections between the literature and our proofs. The first key ingredient of our proofs is a decomposition lemma for graphs avoiding an ‘attached model’ of a fixed graph. This argument is inspired by a result of Kawarabayashi [17] on rooted minors, which in turn is inspired by results of Robertson and Seymour [24]. The proof of Theorem 2 is an inductive argument that uses a result of Illingworth, Scott and Wood [16] in the base case. The proof of Theorem 3 relies on the same decomposition lemma as the proof of Theorem 1. The ordering of the vertices witnessing the bound on wcol_r in Theorem 3 is built via chordal partitions—a powerful proof technique originally developed by Andreae [1] in the context of the cops and robber game, subsequently rediscovered and used by van den Heuvel, Ossona de Mendez, Quiroz, Rabinovich and Siebertz [15] to bound weak coloring numbers, and has subsequently found several other applications in structural graph theory [26]. Another ingredient here is a variant of the Helly property for K_t -minor-free graphs that is of independent interest. This part builds on the work of Pilipczuk and Siebertz [20] for bounded genus graphs, and on the Graph Minor Structure Theorem of Robertson and Seymour [25] for K_t -minor-free graphs.

References

- [1] Thomas Andreae. On a pursuit game played on graphs for which a minor is excluded. *J. Combin. Theory, Series B*, 41(1):37–47, 1986.
- [2] Prosenjit Bose, Vida Dujmović, Mehrnoosh Javarsineh, and Pat Morin. Asymptotically optimal vertex ranking of planar graphs, 2023. [arXiv:2007.06455](https://arxiv.org/abs/2007.06455).
- [3] Rutger Campbell, Katie Clinch, Marc Distel, J. Pascal Gollin, Kevin Hendrey, Robert Hickingbotham, Tony Huynh, Freddie Illingworth, Youri Tamitegama, Jane Tan, and David R. Wood. Product structure of graph classes with bounded treewidth, 2022. *Combin. Probab. Comput.* to appear, [arXiv:2206.02395](https://arxiv.org/abs/2206.02395).

- [4] Chandra Chekuri and Julia Chuzhoy. Polynomial bounds for the grid-minor theorem. *J. ACM*, 63(5), 2016. [arXiv:1305.6577](#).
- [5] Julia Chuzhoy and Zihan Tan. Towards tight(er) bounds for the excluded grid theorem. *J. Combin. Theory, Series B*, 146:219–265, 2021. [arXiv:1901.07944](#).
- [6] Marc Distel, Robert Hickingbotham, Tony Huynh, and David R. Wood. Improved product structure for graphs on surfaces. *Discrete Math. Theor. Comput. Sci.*, 24(2):#6, 2022. [arXiv:2112.10025](#).
- [7] Vida Dujmović, Louis Esperet, Cyril Gavoille, Gwenaël Joret, Piotr Micek, and Pat Morin. Adjacency labelling for planar graphs (and beyond). *J. ACM*, 68(6):42, 2021. [arXiv:2003.04280](#).
- [8] Vida Dujmović, Gwenaël Joret, Piotr Micek, Pat Morin, Torsten Ueckerdt, and David R. Wood. Planar graphs have bounded queue-number. *J. ACM*, 67(4), 2020. [arXiv:1904.04791](#).
- [9] Vida Dujmović, Louis Esperet, Gwenaël Joret, Bartosz Walczak, and David R. Wood. Planar graphs have bounded nonrepetitive chromatic number. *Advances in Combinatorics*, #5, 2020. [arXiv:1904.05269](#).
- [10] Vida Dujmović, Robert Hickingbotham, Gwenaël Joret, Piotr Micek, Pat Morin, and David R. Wood. The excluded tree minor theorem revisited, 2023. *Combin. Probab. Comput.* to appear, [arXiv:2303.14970](#).
- [11] Zdeněk Dvořák. Constant-factor approximation of the domination number in sparse graphs. *European J. Combin.*, 34(5):833–840, 2013. [arXiv:1110.5190](#).
- [12] Michał Dębski, Stefan Felsner, Piotr Micek, and Felix Schröder. Improved bounds for centered colorings. *Advances in Combinatorics*, #8, 2021. [arXiv:1907.04586](#).
- [13] Louis Esperet, Gwenaël Joret, and Pat Morin. Sparse universal graphs for planarity. *J. London Math. Soc.*, to appear. [arXiv:2010.05779](#).
- [14] Martin Grohe, Stephan Kreutzer, Roman Rabinovich, Sebastian Siebertz, and Konstantinos Stavropoulos. Coloring and covering nowhere dense graphs. *SIAM J. Discrete Math.*, 32(4):2467–2481, 2018. [arXiv:1602.05926](#).
- [15] Jan van den Heuvel, Patrice Ossona de Mendez, Daniel Quiroz, Roman Rabinovich, and Sebastian Siebertz. On the generalised colouring numbers of graphs that exclude a fixed minor. *European J. Combin.*, 66:129–144, 2017. [arXiv:1602.09052](#).
- [16] Freddie Illingworth, Alex Scott, and David R. Wood. Product structure of graphs with an excluded minor, 2022. [arXiv:2104.06627](#).
- [17] Ken-ichi Kawarabayashi. Rooted minor problems in highly connected graphs. *Discrete Math.*, 287(1-3):121–123, 2004.
- [18] Jaroslav Nešetřil and Patrice Ossona de Mendez. *Sparsity - Graphs, Structures, and Algorithms*, volume 28 of *Algorithms and combinatorics*. Springer, 2012.
- [19] Marcin Pilipczuk, Michał Pilipczuk, and Sebastian Siebertz. Lecture notes for the course “Sparsity” given at Faculty of Mathematics, Informatics, and Mechanics of the University of Warsaw, Winter semesters 2017/18 and 2019/20. <https://www.mimuw.edu.pl/~mp248287/sparsity2>.
- [20] Michał Pilipczuk and Sebastian Siebertz. Polynomial bounds for centered colorings on proper minor-closed graph classes. *J. Combin. Theory, Series B*, 151:111–147, 2021. [arXiv:1807.03683](#).
- [21] Michał Pilipczuk and Marcin Wrochna. On space efficiency of algorithms working on structural decompositions of graphs. *ACM Trans. Computation Theory*, 9(4), 2018. [arXiv:1509.05896](#).
- [22] Neil Robertson and Paul Seymour. Graph minors. I. Excluding a forest. *J. Combin. Theory, Series B*, 35(1):39–61, 1983.
- [23] Neil Robertson and Paul Seymour. Graph minors. V. Excluding a planar graph. *J. Combin. Theory, Series B*, 41(1):92–114, 1986.
- [24] Neil Robertson and Paul Seymour. Graph minors. XIII. The disjoint paths problem. *J. Combin. Theory, Series B*, 63(1):65–110, 1995.
- [25] Neil Robertson and Paul Seymour. Graph minors. XVI. Excluding a non-planar graph. *J. Combin. Theory, Series B*, 89(1):43–76, 2003.
- [26] Jan van den Heuvel and David R. Wood. Improper colourings inspired by Hadwiger’s conjecture. *J. London Math. Soc.*, 98:129–148, 2018. [arXiv:1704.06536](#).